

Sanjay Vijay

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EDUCATION

Purdue University

Aug. 2024 - May 2027

Bachelor of Science in Computer Science (GPA: 3.98)

West Lafayette, IN

- Concentration in Machine Intelligence, Concentration in Algorithms, Minor in Mathematics, Minor in Statistics
- Relevant Coursework: Competitive Programming, Data Structures and Algorithms, (Spring 2026) Artificial Intelligence

EXPERIENCE

Undergraduate Researcher

Dec. 2025 - Present

Cognitive Robot Autonomy and Learning (CoRAL) Lab

West Lafayette, IN

- Researched techniques for 3D mapping from images captured by robots in ultra-low-light environments
- Verified results from Gaussian Splatting, InstantSplat, and MAST3R algorithms, contributing to in-progress paper draft

Algorithm Subteam Lead

Aug. 2024 - Present

Purdue RoboMaster

West Lafayette, IN

- Led onboarding workshops to teach 30 incoming members computer vision, navigation, and ROS for competition
- Designed pipelines for image segmentation, automatic aim, and ballistic estimation, using OpenCV, ROS 2, and C++
- Enhanced planners and controllers for autonomous LiDAR navigation, completely preventing obstacle collisions

Software Engineering Intern

May 2025 - July 2025

Draper

Cambridge, MA

- Developed a post-mission analysis library for autonomous underwater vehicles (AUVs) used internally by 20 engineers
- Automated interactive visualizations of simulation and in-water data from AUVs, using Plotly, Dash, and C++
- Led transition of AUV data analysis utilities from MATLAB to Python, completely eliminating licensing costs

Mathematics Tutor

Jan. 2025 - May 2025

Purdue University

West Lafayette, IN

- Tutored undergraduate students in important introductory math courses including Calculus I and Calculus II
- Supported students with homework help, exam preparation, and early head starts on challenging course content

Software Engineering Intern

June 2024 - Aug. 2024

Draper

Cambridge, MA

- Collaborated with 8 engineers using Agile methodologies to create and demonstrate teleoperated micro rover prototype
- Developed entire rover codebase with Python libraries to interface with I2C devices over Raspberry Pi 5 GPIO
- Developed tracking system with Kalman filters, Multiple Hypothesis Tracking, and Interacting Multiple Models (IMM)
- Researched sensor fusion literature to implement IMM with MATLAB, enabling robustness to shifting motion patterns

PROJECTS

Neurosphere

Apr. 2025

- Collaborated to develop website that classifies and locates brain tumors from MRI scans uploaded by users
- Applied transfer learning to train a classification model using PyTorch, achieving 98% test accuracy
- Researched and implemented the Grad-CAM algorithm to locate tumors without position labels during training
- Won award for “Best Use of MongoDB” at HackTech, Caltech’s annual hackathon with over 400 participants

SKILLS

Programming Languages - Python, C/C++, Java, MATLAB, R

AI & Robotics - PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, ROS 2, Nav2

Data Science - NumPy, SciPy, Pandas, Matplotlib, Seaborn, Plotly, Dash

Workflow & Development - Agile, Kanban, Jira, Confluence, Notion, Git, Linux